

Predicting local DED track width from thermal history

Audited physical alignment, local boundaries, uncertainty, and held-out cross-condition evaluation.

Executive result

On untouched Track 21, the audited thermal-history Ridge model reduces local-width MAE by 12.3% relative to the original notebook Gradient Boosting baseline retrained under the same split. It is a better benchmark, not a control-ready model: held-out R^2 remains negative and the nominal 90% interval under-covers.

0.139 mm	12.3%	0.088 mm	76.5%
Held-out MAE	MAE reduction	Grouped CV MAE	Interval coverage

Model	MAE	RMSE	R^2	Coverage
Notebook Gradient Boosting	0.159 mm	0.187 mm	-0.99	81.5%
Audited Ridge alpha=10	0.139 mm	0.167 mm	-0.58	76.5%

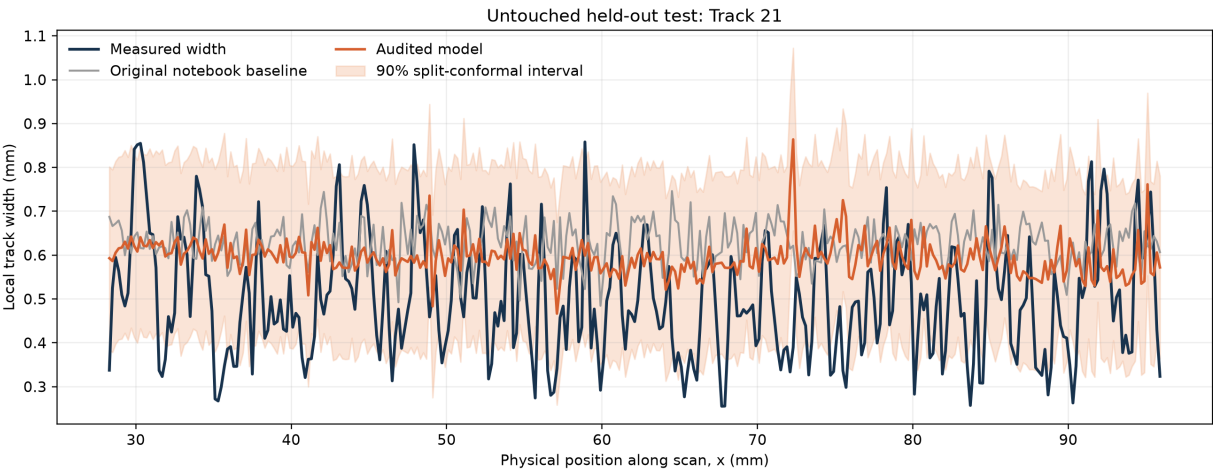


Figure 1. Measured local width, reproduced notebook baseline, audited prediction, and development-calibrated interval on held-out Track 21.

Method: local geometry without label leakage

Target and alignment. Each profilometer cross-section is robustly detrended. Connected 30%-of-peak crossings around the central bead maximum define left and right boundaries; width is their difference. Invalid acquisition gaps remain excluded. Thermal frames map to physical x using 10 mm/s at 50 fps (0.2 mm/frame).

Thermal history. Features include hot-region area, bounding-box geometry, equivalent diameter, elongation, temperature percentiles, hot-region mean, thermal mass, gradients, asymmetry, deltas, one-frame lags, three-frame rolling means, and low-order physical-x harmonics.

Audit item	Prior artifact	Audited protocol
Target	Height/average language	Spatial width plus both boundaries
Alignment	Normalized row position	Physical x in millimeters
Selection	Random split / test reused	Grouped CV on Tracks 8, 10, 14
Final test	No untouched condition	Track 21 opened once
SEM	Assumed dominant	Masked ablation; rejected when CV worsened
Uncertainty	Implied reliability	Conformal interval plus observed coverage

Selection and interpretation

Ridge $\alpha=10$ with thermal-only features minimizes leave-one-track-out development MAE (0.0876 mm). The best thermal-plus-masked-SEM candidate reaches 0.1135 mm, so SEM is not selected. With only four conditions, process and substrate effects are not identifiable; the report makes no causal substrate claim.

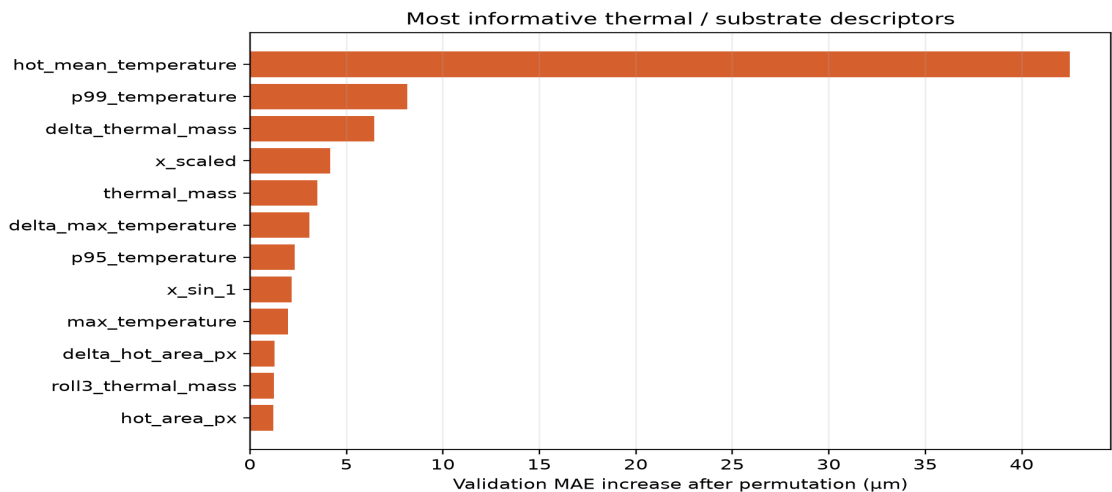


Figure 2. Development-validation permutation importance. Hot-region temperature and thermal mass lead; importance indicates association, not causality.

Boundary results, uncertainty, and release gates

Held-out signal	MAE	RMSE	R ²
Left boundary	0.151 mm	0.192 mm	-0.15
Right boundary	0.197 mm	0.251 mm	-0.12
Mean boundary MAE	0.174 mm	-	-

What is supported

- Held-out width MAE and RMSE improve under a strict, same-sample comparison.
- Short thermal history generalizes better than masked SEM on this split.
- Connected boundary extraction produces explicit local left/right predictions.

What remains open

- Held-out R² is -0.58; the unseen-condition shift still dominates local fit.
- Nominal 90% intervals cover 76.5% of Track 21, below the target.
- Track 21 profilometry is incomplete; only valid aligned regions are scored.
- Four conditions cannot separate substrate-driven from process-driven variation.

Release gate: do not use this model for closed-loop control until interval coverage and positive held-out R² are demonstrated across additional powers, plates, and repeats.

Reproduce

```
python scripts/run_final_analysis.py --raw-dir /path/to/data --output-dir
results/final_submission
python -m pytest
python scripts/build_final_report.py
```

Generative AI disclosure

OpenAI Codex assisted with code review, test generation, debugging, and artifact layout. Reported metrics were produced by the tracked analysis code and verified against saved predictions; AI did not supply or alter experimental measurements.

Sources and submission record

Dataset: <https://doi.org/10.5281/zenodo.21285367>

Paper: <https://arxiv.org/abs/2607.07965>

Locked metrics: results/final_submission/metrics.json

Executed notebook: notebooks/03_final_submission_audited.ipynb

Bottom line: the audited model is measurably better than the reproduced notebook baseline while the negative R² and interval under-coverage keep the claim honest.